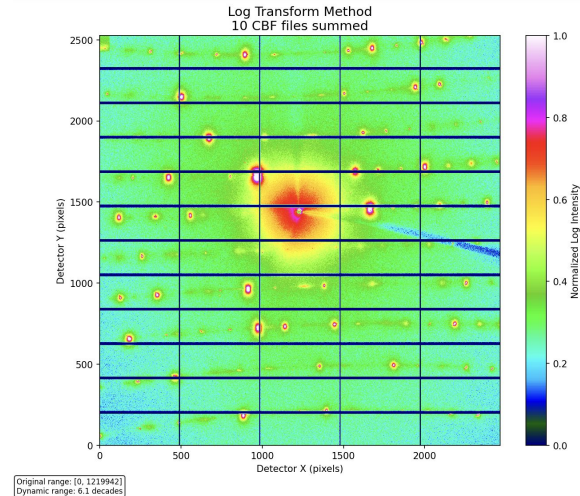
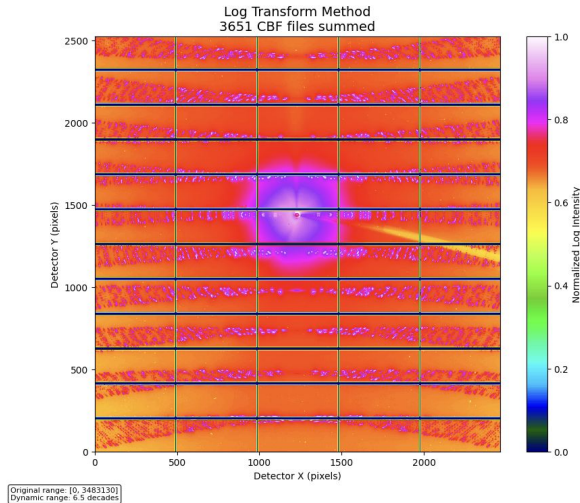


Data, Statistics, and Difference Imaging

Week 4

Data Collection

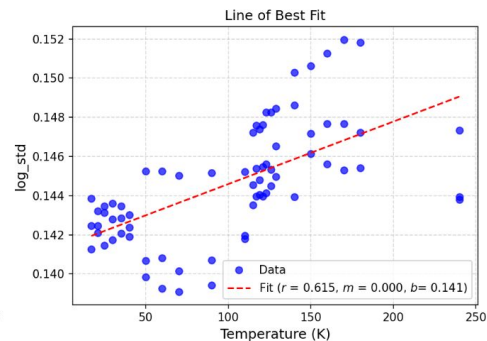
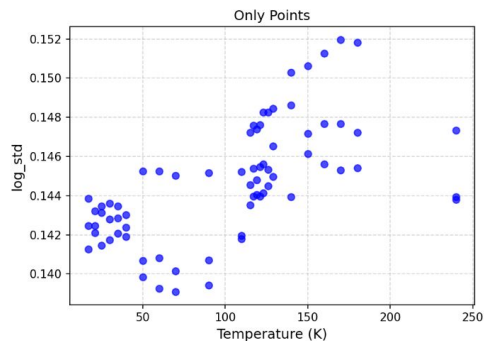
- Full collection at one temperature: ~6 minutes for the 3650 in 1 angular position, ~20 minutes to get through all 3
- Temperature change: ~20-30 minutes for 5K
- Primary Goal: analyze using first 10 images (<1 minute collection) to determine whether transition occurring/occured
 - Reduce data collection by ~95% per temperature, notable improvement
- Secondary/Tertiary Goals: Look at peak shifts and possible diffuse scattering



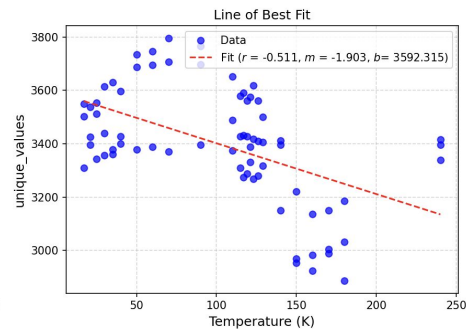
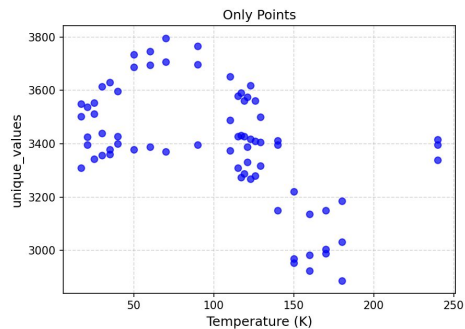
Statistics

- Used provided load_cbf.py
- Graphed various statistics versus temperature
- Some seemingly had a linear correlation
 - Possible its real or possible its a factor of small data
 - Also possible its detector artifacts

Temperature dependence of log_std

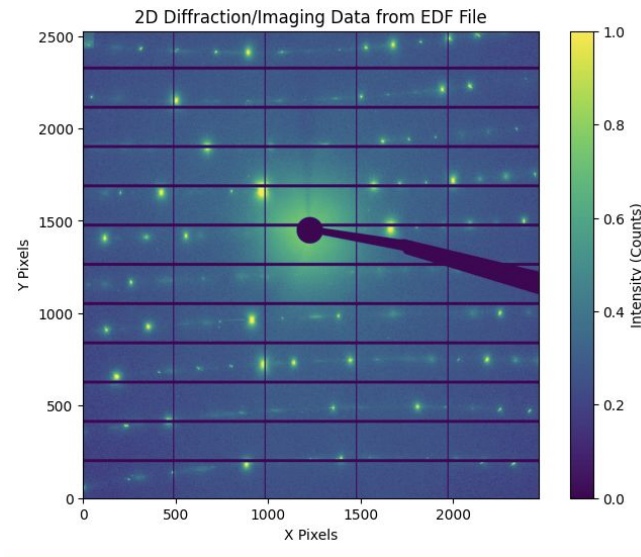


Temperature dependence of unique_values



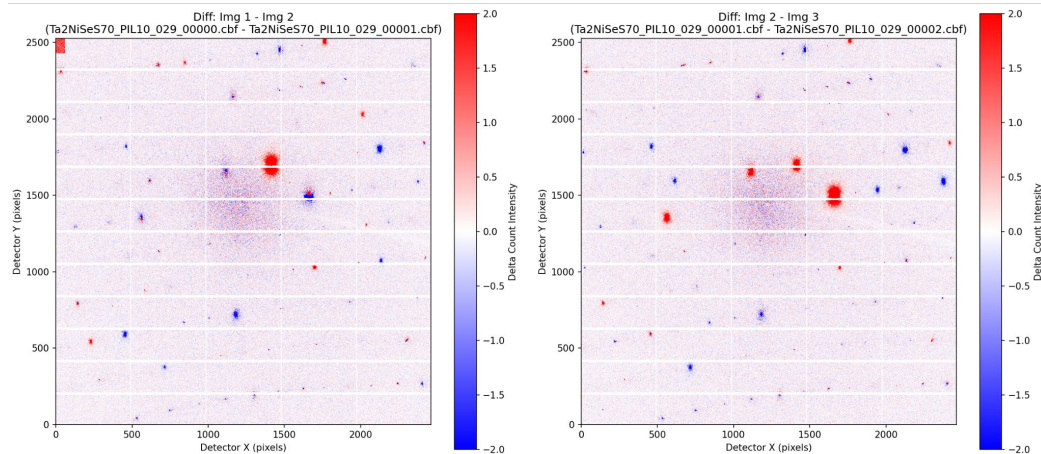
Data

- Mask tested to see effectiveness
- Mask does not account for notable halo effect of central region
 - Unrelated, but weird top-left artifact
- Possible future work to fit 2D Gaussian to central area
- Other possible future work to fit Gaussian to overall peaks
 - Check whether PSF for the detector already exists
 - Would be important for looking diffuse scattering



Difference Imaging

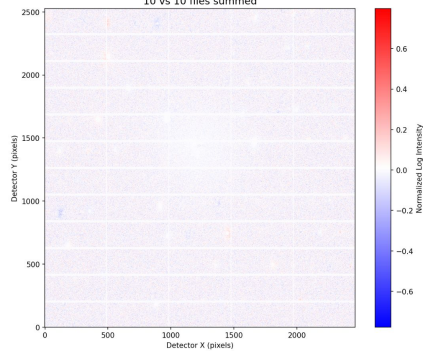
- Difference imaging conducted in 2 ways
- First way consecutive image subtraction
 - Seems to imply notable changes image-to-image
 - May be due to noise or detector artifacts, possible doesn't happen for other temperatures/images



Difference Imaging pt. 2

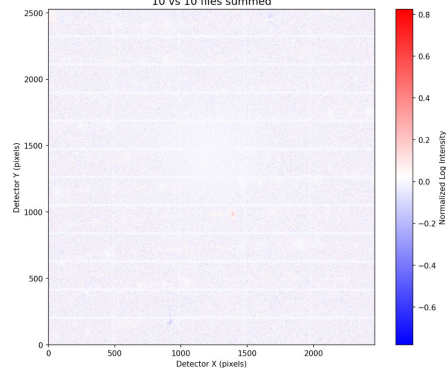
- Second way difference imaging between summed at different temperatures
 - 10 images summed at each temperature, normalized the same, and subtracted
 - Worked quite well, though tbd how exactly to confirm

Diff: /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSe570/S1/50/Ta2NiSe570_024 - /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSe570/S1/60/Ta2NiSe570_027
10 vs 10 files summed



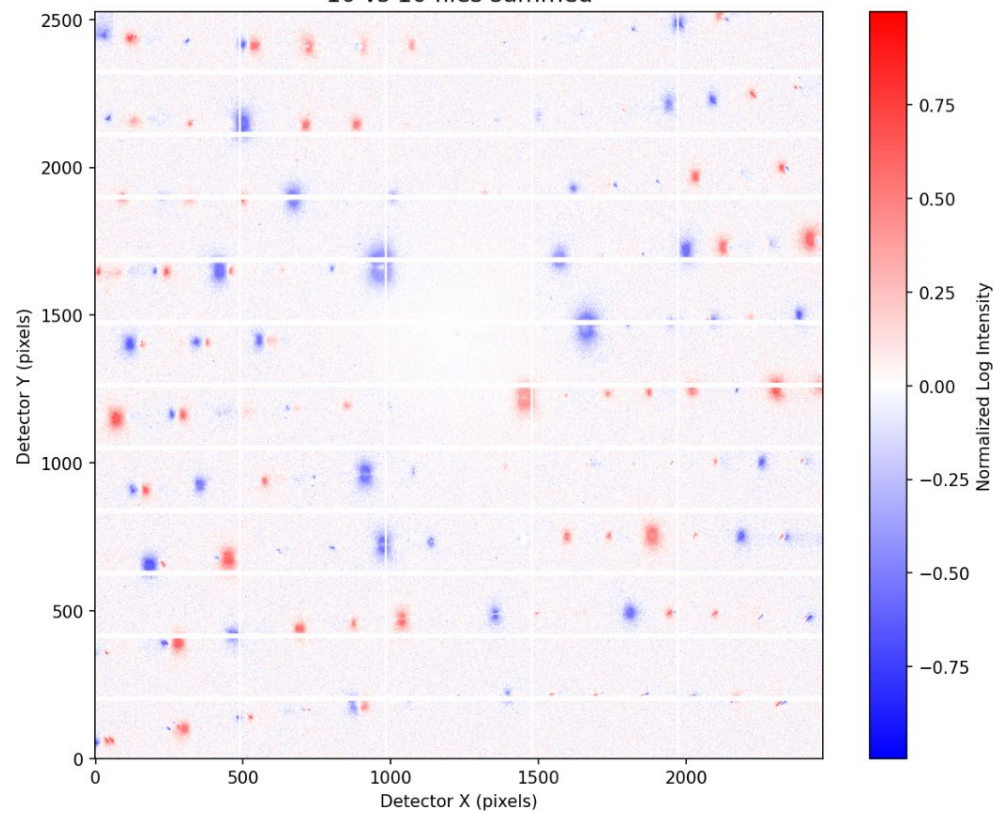
0.17354254678114356

Diff: /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSe570/S1/35/Ta2NiSe570_018 - /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSe570/S1/40/Ta2NiSe570_021
10 vs 10 files summed



0.17404240755470998

Diff: /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSeS70/S1/40/Ta2NiSeS70_021 - /nfs/chess/id4b/2025-2/he-4025-d/raw6M/Ta2NiSeS70/S1/50/Ta2NiSeS70_024
10 vs 10 files summed



0.1838485925186182